

Section 2. Highlights: Acne Pathogenesis and Role of Diet

Key Points

- Acne originates in the **pilosebaceous unit**, primarily affecting areas with high density of these units such as the **face, neck, and upper torso**
- **Inflammation** is a central driver in acne development, present from the earliest preclinical stages through to lesion formation and scarring
- The **pathophysiology of acne is multifactorial**, involving: - Increased **sebum production** - **Hyperproliferation of keratinocytes** within the sebaceous follicle - **Follicular hyperkeratinization** - **Colonization by *Cutibacterium acnes*** (formerly *Propionibacterium acnes*) - **Inflammatory responses**

Acne Pathophysiology

Acne is a complex, multifactorial disorder centered on the pilosebaceous unit.^{1,27,28} Advances in molecular research have revealed that inflammation is triggered when growth factors and androgens activate the pilosebaceous unit. Inflammation plays a critical role at every stage of acne—beginning with the formation of the **microcomedone**, the precursor to visible lesions, and extending to long-term consequences such as **post-inflammatory hyperpigmentation and scarring**.^{29–32}

This inflammatory process occurs concurrently with: - **Excess sebum production** - **Abnormal keratinocyte proliferation and differentiation** - Colonization and amplification of inflammation by *Cutibacterium acnes* within the follicular duct, particularly in the infra-infundibulum (Fig. 3).^{28}

Figure 4 summarizes the proposed sequence of events in acne development, although the exact temporal order remains incompletely defined. ^{28} Current evidence suggests that **insulin-like growth factor-1 (IGF-1)** and **androgens** (e.g., DHEA-S) act synergistically—especially during puberty—to stimulate sebaceous gland activity. ^{28,33} These hormonal signals induce both increased sebum volume and qualitative changes in sebum composition, including: - ↓ Linoleic acid levels (due to sebaceous gland hyperplasia) - ↑ Squalene and lipid peroxides - ↑ Ratio of saturated to monounsaturated fatty acids ^{33–36}

Multiple additional factors influence sebum regulation, including environmental exposures, neuroendocrine signals, and inflammatory mediators. Emerging data indicate that **endocannabinoids**, **ectopeptidases**, **corticotropin-releasing hormone (CRH)**, **α-MSH**, **histamine**, **acetylcholine**, and others bind to receptors on sebocytes and may modulate their function in acne-prone skin. ^{37}

Structural differences are also observed in acne-prone individuals: - Greater number of **lobules per sebaceous gland** - Larger **follicular diameter** ^{38} - Thicker, denser **follicular corneocyte lamellae** - Reduced single-cell desquamation, promoting follicular occlusion ^{38}

Additionally, sebocytes on the face exhibit a heightened response to androgens compared to those on non-acne-prone areas like the legs, which may explain the typical facial and truncal distribution of acne. ^{39}

Androgens and growth factors also stimulate follicular keratinocytes, leading to: - **Hyperproliferation** - Altered differentiation - Shift from individual cell shedding to cohesive clumping at the follicular orifice

Histologically, affected follicular walls show increased: - Lipid vacuoles - Filaggrin - Tonofilaments

The pilosebaceous unit—particularly sebocytes—plays an active role in **skin endocrine function**. ^{40} Sebaceous glands secrete sebum via **holocrine secretion**, where mature sebocytes undergo programmed disintegration

involving DNA degradation before releasing lipids into the sebaceous duct. This process is distinct from apoptosis or necroptosis. ^{41}

Sebaceous gland activity is regulated by: - **Androgen hormones** - **Growth-promoting hormones** - **Growth factors** (e.g., IGF-1)